

AI Engineering Learning Resources (2026)

1. Python Programming

Resource	Type	Link
Harvard CS50P: Introduction to Programming with Python	Free Course	https://cs50.harvard.edu/python/
Kaggle - Python Course	Free Course	https://www.kaggle.com/learn/python
freeCodeCamp - Scientific Computing with Python	Free Course	https://www.freecodecamp.org/learn/scientific-computing-with-python/

2. Mathematics for AI

Resource	Type	Link
Khan Academy - Linear Algebra	Free Course	https://www.khanacademy.org/math/linear-algebra
Khan Academy - Statistics & Probability	Free Course	https://www.khanacademy.org/math/statistics-probability
3Blue1Brown - Essence of Linear Algebra	YouTube Playlist	https://www.youtube.com/playlist?list=PLZHQObOWTQDMsr9K-rj53DwVRMYO3t5Yr
3Blue1Brown - Calculus Series	YouTube Playlist	https://www.youtube.com/@3blue1brown

3. Machine Learning

Resource	Type	Link
Machine Learning Specialization (Andrew Ng)	Course	https://www.coursera.org/specializations/machine-learning-introduction
Google Machine Learning Crash Course	Free Course	https://developers.google.com/machine-learning/crash-course
Kaggle - Intro to Machine Learning	Free Course	https://www.kaggle.com/learn/intro-to-machine-learning

Resource	Type	Link
Kaggle - Intermediate Machine Learning	Free Course	https://www.kaggle.com/learn/intermediate-machine-learning

4. Deep Learning

Resource	Type	Link
Deep Learning Specialization	Course	https://www.coursera.org/specializations/deep-learning
fast.ai - Practical Deep Learning for Coders	Free Course	https://course.fast.ai
PyTorch Tutorials	Documentation	https://pytorch.org/tutorials/
TensorFlow Tutorials	Documentation	https://www.tensorflow.org/tutorials

5. Computer Vision

Resource	Type	Link
OpenCV Official Tutorials	Documentation	https://docs.opencv.org/
Ultralytics YOLO Documentation	Documentation	https://docs.ultralytics.com/
PyTorch Vision Tutorials	Documentation	https://pytorch.org/tutorials/

6. Natural Language Processing (NLP)

Resource	Type	Link
Hugging Face NLP Course	Free Course	https://huggingface.co/learn/nlp-course
spaCy Course	Free Course	https://course.spacy.io
NLTK Documentation	Documentation	https://www.nltk.org/

7. Generative AI & LLMs

Resource	Type	Link
OpenAI Academy	Free Learning Platform	https://academy.openai.com/
DeepLearning.AI Short Courses	Free Courses	https://learn.deeplearning.ai/
Microsoft AI for Beginners	GitHub Course	https://github.com/microsoft/AI-For-Beginners
Hugging Face Learn	Free Courses	https://huggingface.co/learn
Google AI Learning Path	Free Learning	https://grow.google/ai
IBM SkillsBuild AI Courses	Free Courses	https://skillsbuild.org/
Anthropic Learning	Free Resources	https://anthropic.skilljar.com/

8. RAG, AI Agents & LLM Frameworks

Resource	Type	Link
LangChain Documentation	Documentation	https://python.langchain.com/docs/
LangGraph Documentation	Documentation	https://langchain-ai.github.io/langgraph/
LlamaIndex Documentation	Documentation	https://docs.llamaindex.ai/
OpenAI Cookbook	GitHub	https://github.com/openai/openai-cookbook

9. MLOps

Resource	Type	Link
MLflow Documentation	Documentation	https://mlflow.org/docs/latest/
Docker Documentation	Documentation	https://docs.docker.com/
Kubernetes Documentation	Documentation	https://kubernetes.io/docs/
GitHub Actions Documentation	Documentation	https://docs.github.com/actions

10. Cloud Computing

Resource	Type	Link
AWS Skill Builder	Free Courses	https://skillbuilder.aws/
Microsoft Learn	Free Courses	https://learn.microsoft.com/training/
Google Cloud Skills Boost	Free Courses	https://www.cloudskillsboost.google/

Best YouTube Channels

AI & Machine Learning

- Andrew Ng — <https://www.youtube.com/@Deeplearningai>
 - DeepLearning.AI — <https://www.youtube.com/@DeeplearningAI>
 - StatQuest with Josh Starmer — <https://www.youtube.com/@statquest>
 - Yannic Kilcher — <https://www.youtube.com/@YannicKilcher>
 - IBM Technology — <https://www.youtube.com/@IBMTechology>
 - AssemblyAI — <https://www.youtube.com/@AssemblyAI>
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Generative AI

- Krish Naik — <https://www.youtube.com/@krishnaik06>
 - CampusX — <https://www.youtube.com/@campusx-official>
 - freeCodeCamp.org — <https://www.youtube.com/@freecodecamp>
 - Codebasics — <https://www.youtube.com/@codebasics>
 - Cole Medin — <https://www.youtube.com/@ColeMedin>
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Python

- Corey Schafer — <https://www.youtube.com/@coreyms>
 - Programming with Mosh — <https://www.youtube.com/@programmingwithmosh>
 - Tech With Tim — <https://www.youtube.com/@TechWithTim>
 - Bro Code — <https://www.youtube.com/@BroCodez>
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Practice Platforms

Platform	Link
Kaggle	https://www.kaggle.com/
LeetCode	https://leetcode.com/
HackerRank	https://www.hackerrank.com/
DrivenData	https://www.drivendata.org/
Papers With Code	https://paperswithcode.com/

Books

- Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow
- Deep Learning (Ian Goodfellow)
- Dive into Deep Learning (<https://d2l.ai/>)
- Clean Code
- Designing Data-Intensive Applications

Research & News

Resource	Link
arXiv	https://arxiv.org/
Papers With Code	https://paperswithcode.com/
Hugging Face Papers	https://huggingface.co/papers
OpenAI Research	https://openai.com/research
Google Research	https://research.google/

Recommended Learning Order

1. Python
2. Mathematics
3. Machine Learning
4. Deep Learning
5. Computer Vision

6. Natural Language Processing
7. Generative AI & LLMs
8. RAG & AI Agents
9. MLOps
10. Cloud Computing
11. Build Production-Ready Projects